NATIONAL UNIVERSITY
of Computer & Emerging Sciences, Lahore

Department of Computer Science

CS118 – Programming Fundamentals
FALL 2024

Instructor Name: Dr. Aamir Wali **TA Name:**
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Office Location/Number: New Block
Office Hours: **Tue & Thu 10 am – 11 am**

Course Information

Program: BS (CS,DS,SE) **Credit Hours:** 3 + 1 (Lab) **Course Type:** Core

Course Description/Objectives/Goals:

- To introduce the notion of algorithms.
- To develop problem solving and logic building skills in students.
- To introduce the basic concepts of programming in C++, including basic data types, expressions, iterations, functions and arrays.

Course Learning Outcomes (CLOs):

At the end of the course students will be able to:	**Domain**	**BT* Level**
Understand basic problem solving steps and logic constructs	C	2
Apply basic programming concepts	C	3
Design and implement algorithms to solve real world problems and should be able to translate a problem statement into pseudo-code/C++ code	C	3
* BT= Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A= Affective domain **Bloom's taxonomy Levels:** 1. Knowledge, 2. Comprehension, 3. Application, 4. Analysis, 5. Synthesis, 6. Evaluation		

Course Textbook

1. C++ Programming: Program Design Including Data Structures, by D. S. Malik (8th Edition)
2. C++: How to Program? by Deitel & Deitel (10th Edition)

Additional references and books related to the course:

1. Theory and Problems of Programming with C++ by John R. Hubbard, 2nd Edition
2. Programming and Problem Solving with C++, Nell Dale
3. www.learncpp.com

Tentative Weekly Schedule

Week	Session
1	Intro to computers, what is computer programming? 1. Pseudo code notations 2. Developing simple programs 3. Memory concepts
2	1. More on problem solving, components of an algorithm. Define and develop simple algorithms
3	1. Explain the structure of C/C++ program. 2. Define source code, object code and Machine code How to deal with a language editor, compile and run the c / c++ simple file. Different Component of a Program 1. Explain the purpose of main function in C++ program. 2. Write the single line and multiple line comments in the source file. Importance of Variables 1. Declaring and Initializing Variables. 2. Primitive data types (int, float, chat, double) 3. Statement and Expressions 4. Operators 1)
4	1. Clean code (indentation) 2. Operator precedence 3. Type casting Different Flows Of “Control” In A Program 1. Simple and compound statements 2. Sequential logic 3. Selection logic 1) The if / else statement 2) else if construction 3) switch statement (optional)
5	Different Flows Of “Control” In A Program 4. Iterative logic 1) while statement 2) do – while statement 3) for statement 5. Exceptions in control structures 1) continue statement 2) break statement 3) return statement 4) exit function (exit from main)
MID-I	
6	Different Flows Of “Control” In A Program

	6. Nested loop structures

7	Single and multi-dimensional arrays
	
	1. Purpose
	2. declaration
	3. definition
	4. initialization through initializer
	5. accessing array elements
	6. array organization in memory and element access using Array name and index.
	7. Initialization using member initializer list, and by using loops.
	8. searching elements in arrays, through
	    1) linear search (coding)
	    2) binary search (coding + algorithm analysis in terms of efficiency)

8	Single and multi-dimensional arrays
	
	9. sorting of arrays
	    1) bubble sort (coding + analysis)
	    2) quick sort (introduction)
	10. merge sorting arrays
	11. find Min, Max, Avg, Equilibrium Index
	12. reverse: All Elements, odd/eve elements and indices
	13. character arrays as strings
	14. useful functions of String.H
	15. useful functions of CType.H
	
	arrays of strings
	
	1. Difference between Null terminated CStrings and character arrays.
	2. Storage of CStrings in character arrays and aggregate I/O.
	3. Find String length, Compare strings,
	4. Find substring and replace,
	5. Calculate frequency of specific characters Remove specific characters.

9	2-Dimensional Array 1. 2-D Array, how it is organized in memory in row/col major order. 2. Initialization using member initializer list, and by using loops. 3. I/O and processing of elements in row/col major order. 4. Application: Store and process Students Quiz marks. 5. Find Min, Max, Avg, column and row wise. 6. Sorting: row wise or column wise, complete array by specific column or row.
10	Application: 1. Sets, Union, Intersection, difference. 2. Shifting and Rotation of elements: right and left 3. Insert and delete elements from ordered list using shifting 4. Application: Matrices storage and processing 5. Addition, Subtraction, Multiplication, 6. Transpose, 7. Check for Upper and lower triangular.
11	6. Pointers

MID-II	

12	Use of Functions in C / C++
	
	1. Identify the components of a function.
	    1) Function prototype
	    2) Function definition
	    3) Signature vs. return type
	    4) Parameter vs. argument
	2. functions design
	3. making a function call
	4. functions pass by value vs. pass by reference

13	Use of Functions in C / C++
	
	5. function types
	    1) Recursive function
	    2) Overloaded function

14	Use of Functions in C / C++ 1. passing arrays to functions 2. returning array from a function (static arrays only) 3. design different functions for input, output, search, reverse 4. passing 2D arrays to functions: Complete, individual rows, or elements. 5. processing diagonals: reverse elements, print data of whole array. Use of graphic libraries functions.
15	File handling 1. I/O from simple text Files in arrays. Review

(Tentative) Grading Criteria:

1. Assignments + Homework **(10 %)**
2. Quizzes **(10 %)**
3. Midterms **(30 %)**
4. Project **(10 %)**
5. Final Exam **(40 %)**

- Grading scheme for this course is **Absolute** under application of CS department's grading policies.
- Minimum requirement to pass this course is to obtain at least **50%** absolute marks

Course Policies:

- All assignments and homework must be done individually.
- Late Submissions of assignments will not be accepted.
- **Plagiarism** in any work (Quiz, Assignment, Midterms, Project and Final Exam) from any source, Internet or a Student will result in **deduction of absolute marks or F** grade.
- Minimum **80%** attendance is required for appearing in the Final exams.